

The Next Step for v16.0: Material Gaps against v11.0

An archaeology review — TuringAward

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1 Method

I inventoried the whole v11.0 corpus and classified each material item against the v16 artifact. **v11.0 read:** the unified spec body (`ledger/drafts/sec01-20`, `appA-I`, via the file set and the signed `CHANGELOG_v10.3_to_v11.0.md`); `deferredSettlement.tex` (2129 lines) with its Pareto declaration; `market_data_mvp_v1.0/v2.0.tex`; `data/ledger_data_v1.0.tex`; `valuation/ledger_valuation_v1.0.tex` the convergence and sign-off memos. **v16 artifact:** `Ledger_Spec_v16.0/ledger/ledger_v16_0.tex` — the single consolidated, certified file (110pp). It is the only v16 candidate in the repository, so no panel ruling on artifact identity was needed; the identification was unambiguous and is recorded here as required. Each v11 item was marked PRESENT (cited in v16), TRANSFORMED (cited, with what changed), or ABSENT. Every ABSENT item was then subclassified MISSING (valuable, no evidence of deliberate removal) versus RETIRED, using the v15-era `EXCLUSIONS.md` register (E1–E76), the named companions in v16 Ch. 17 (Worked-Examples, SBL-Operations, Track-B, Managed-Account, Simulation), and the out-of-scope list as the deliberate-retirement trail. RETIRED items are dropped unless their retirement is argued a mistake; none met that bar. *Finding:* *v16 is a remarkably complete absorption of v11.* The vast majority of v11 is PRESENT or TRANSFORMED — the three-home model, the futures variation-margin identity, the coordinate vector and coverage invariant, corporate actions and the market-data operator, close-out netting, the settlement-obligation unit with its partial-edge split and cum/ex market-claim, virtual ledgers and the TRS, bitemporal time travel, and an executable-check regime with firing witnesses. Five gaps survived the filter.

2 Material Gaps

NS-01 — The settlement–custody reconciliation identity is gone. v11’s deferred-settlement spec carried an explicit algebraic identity binding the ledger’s internal balances to the external nostro/depot books across the open settlement window: $\text{nostro}_{\text{ext}} = \text{own}(\text{ccy}) + \sum_{\text{cpty}} w_{PS}[\cdot] - \text{inflight}_{\text{out}} + \text{inflight}_{\text{in}}$, with a depot analogue, both computable in time bounded by the wallet count rather than the transaction count (`deferredSettlement.tex` §8.1–8.5, invariant DS3; the Conservation Lifting Theorem §8.8 makes the identity a corollary). The same identity appears independently in the SBL chapter as the custodian check $\text{own} + \text{borr} + \text{inflight} = \text{depot}$ (`sec16`, L466), and appendix C lists nostro/vostro and custodian/depot breaks as failures the design detects. *Absent in v16:* Ch. 11 re-books **owned** at instruction and carries the instruction-to-settlement gap in the settlement-obligation unit, but states no reconciliation identity to the custodian; “reconcile against the custodian” is named at the boundary and never made computable (the closest thing, dual valuation’s per-venue price check, reconciles *value*, not the position/cash balances). This is the one internal–external check the constitution keeps *in* scope (reconciled against at the boundary), yet it is the ledger’s only defence against the exact drift that motivated the deferred-settlement work. *Lens:* *L2 (transactions and data — reconciliation).* *Reinstate:* add one subsection to Ch. 11 stating the nostro/depot identity as a projection over **owned** plus the open settlement-obligation units per

counterparty, with the three-way classify (clean / expected lead-lag / break) that distinguishes an open-window state from a true break, and one executable property asserting the identity holds within tolerance on generated settlement histories. *Cost*: ≈ 1 pp, no new primitive (it reads existing coordinates and units). *Risk*: low; the risk of omission is higher — a settlement book with no computable custody check re-admits the reconciliation failure the framework exists to abolish.

NS-02 — No reference currency, FX translation, or currency minor-unit variation.

Four v11 documents carry a currency dimension: the data layer registers currencies with per-currency `minor_units` (USD 2, JPY 0) (`ledger_data_v1.0.tex` L_1); the market-data MVP works a bitemporal EUR/USD restatement to the pip; valuation calibrates FX state (3–5 latent factors); and deferred settlement models a cross-currency trade as a parent FX obligation with two child legs, the open legs a *quantified Herstatt exposure* (`deferredSettlement.tex` §7.4, DS10). *Absent in v16*: the specification assumes a single implicit currency throughout — “exact integers in minor units,” “exact in cents” — with no reference-currency NAV, no FX-rate observation feeding a translation operator, and no cross-currency settlement leg or Herstatt visibility. A book valued and settled in one currency is not the trading book in scope. The “cents” assumption is itself a latent defect: rounding a JPY figure to two decimals is a hundred-fold error. *Lens*: L_9 (*numerical computation — currency arithmetic and minor units*). *Reinstate*: state that money is an integer count at the currency’s declared minor-unit scale (a datum of the currency unit, not a global constant); admit the FX rate as an ordinary recorded observation; make reference-currency NAV an FX-translation projection over the same composition; and carry a cross-currency settlement obligation as two legs whose open state projects as Herstatt exposure. *Cost*: ≈ 1 –1.5 pp across Ch. 7 (valuation) and Ch. 11 (settlement). *Risk*: medium — FX translation touches the valuation and settlement projections, but adds no new primitive (an FX rate is an observation, a translation is pricing).

NS-03 — The client-isolation (segregation) theorem, and its false-theorem warning, were dropped.

The v10.3→v11.0 changelog records a correctness-critical correction: v10.3’s “conservation alone enforces segregation by algebra” was a *false theorem* — it permits a cross-client move — replaced in v11 (`sec09`, `thm:seg`, L_{240}) by segregation as a theorem holding only under $\text{CONS} \wedge \text{LOC} \wedge \text{C4}$ (conservation, locality, and the capability-read condition); the same “conservation alone is insufficient” warning drives the singleton-bilateral mandate invariant (L_{44}). The v15 exclusions register (E_{31}) explicitly promised “the segregation theorem itself is carried.” *Absent in v16*: client isolation *within* a ledger rests on a single sentence — “authorisation is checked against the manager” (Ch. 3, Wallets) — with no theorem, no locality/capability precondition, and no record of the warning that conservation alone is *insufficient*. (v16 does carry the *cross-ledger* realm isolation, v11’s P_7 — “the bridge is an observation, never a move” (Ch. 10) — but that is a different property: it isolates the virtual ledger from the true one, not one client’s wallets from another’s inside one ledger.) The gap is not decorative: it is the property under which one manager cannot move another beneficial owner’s assets. A register that claims an item is carried while the artifact omits it is the strongest possible evidence of a MISSING, not a RETIRED, item. *Lens*: L_1 (*concurrency and isolation — and L_8 capability; the safety property a single writer does not by itself supply*). *Reinstate*: state the isolation theorem in Ch. 6 or Ch. 4 — authorised balance change is confined to wallets the acting manager holds a capability for, so conservation *plus* the capability-read gate *plus* move-locality makes a cross-client move unrepresentable — with an executable property that an unauthorised cross-wallet move is refused. *Cost*: < 0.5 pp. *Risk*: low; it names a guarantee the door already enforces but the prose leaves as an aside.

NS-04 — The recursive partial-settlement split has no termination or fragmentation bound.

v11 bounded partial-fill fragmentation: a partial cascade terminates within $D_{\max} = 2$ else the residual goes Failed→Compensated, with per-step conservation and monotonic advance

(`deferredSettlement.tex` §6.3, invariants DS11a/DS11b). *Diminished in v16*: Ch. 11’s partial edge splits a settlement-obligation unit “again by the same edge, recursively,” with no depth or fragment-count bound stated. It terminates only because quantity is a finite integer, so a Q -unit instruction can fragment into up to Q live obligation units, each entering the obligation-liveness queue — an operational and liveness cost the v11 bound existed to cap. *Lens*: *L3 (specification and verification — liveness and a variant function for termination)*. *Reinstate*: add the fragment bound and its variant to the partial edge (residual strictly decreases; beyond the declared depth the residual is failed-and-compensated rather than re-split), and property-test that no generated settlement history exceeds it. *Cost*: a few lines in Ch. 11 and one property in Ch. 15. *Risk*: low; it strengthens an existing edge and cannot weaken a proved property.

NS-05 — The economic-recomputation oracle does not pin the contract-code version. v16’s economic check re-runs the map on the recorded event and compares: `contract e (ledgerBefore e) == txRecorded` (Ch. 15, `prop_recompute`). This is sound only against the contract as it stood when the event was admitted. v11’s data layer pinned this explicitly: a `VersionPin` with a *contract* axis alongside schema, model, and canonicalisation, so a recomputation years later is deterministic (`ledger_data_v1.0.tex` §5.2). *Absent in v16*: `ProductTerms` are versioned and contracts are called “versioned functions,” but nothing records *which* contract-code version a recomputation must use; an amended contract silently re-derives a different transaction and the oracle either false-fails or false-passes. Replay is safe (it re-reads transactions, never re-executes contracts), but the *economic* oracle re-executes and is therefore under-specified. *Lens*: *L3 (verification — reproducibility of the check itself)*. *Reinstate*: record the contract-code version on the transaction (or bind it to the `ProductTerms` version already recorded) and state that `prop_recompute` reads the contract version in force at that log position. *Cost*: one sentence in Ch. 15 plus a field note in Ch. 5. *Risk*: low; it closes a determinism hole in a check the specification already relies on.

3 Declaration

Lenses applied: L1 (isolation — NS-03), L2 (reconciliation — NS-01), L3 (verification/liveness — NS-04, NS-05), L9 (numerical/currency — NS-02); L8 (capability) considered jointly with NS-03. **Lenses dismissed**: L4 (algorithms/complexity — no hidden hard subproblem; the fold is linear); L5 (abstraction/language — v16’s type discipline is sound and idiomatic); L6 (systems architecture — conceptual integrity is intact, one document, one writer); L7 (networks — no transport is specified; delivery is the untrusted machines’ concern, already handled); L10 (learning — no learned component is load-bearing for any safety property, by design). **Confidence**: MEDIUM-HIGH. The v16 read was complete and the v11 companion inventories were thorough; the residual unknown is whether NS-01 (the custody reconciliation identity) is judged in-scope or an external-authority matter the owner intends to leave at the perimeter — I read it as in-scope because the constitution keeps boundary reconciliation in scope, but that is the one call an owner could decide the other way. **Decision log**: DL-01 | archaeology | which file is the authoritative v16 artifact? | single certified consolidated file present, no rival | ruling: `ledger_v16_0.tex`, unambiguous | no panel convened; no ambiguity met the threshold for a vote.